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ST4005CEM Database System

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1. Create a table called library with the following columns:
Book_id, title, author, price, gener must include Science, History and Biography.

```
1 • create database LMS;
2 • use LMS;
3 • create table library(
4     book_id int primary key auto_increment,
5     title varchar(50),
6     author varchar(50),
7     price float,
8     gener enum("Science","History","Biography")
9 );
10
```

Result Grid

	book_id	title	author	price	gener
*	NULL	NULL	NULL	NULL	NULL

2. Insert the following 10 records into the library table:

```
12 • insert into library(title,author,price,gener)
13 values
14 ("The Alchemist","Paulo Coelho",12.00,"Science"),
15 ("Atomic Habits","James Clear",18.99,"History"),
16 ("1984","George Orwell",10,"Biography"),
17 ("To Kill a Mockingbird","Harper Lee",14.45,"History"),
18 ("The Hobbit","J. R. R. Tolkien",15.00,"History"),
19 ("Rich Dad Poor Dad","Rich Dad Poor Dad",11.2,"Science"),
20 ("The Great Gatsby","F. Scott Fitzgerald",16.44,"Biography"),
21 ("Think and Grow Rich","Napoleon Hill",12.99,"Biography"),
22 ("Harry Potter and the Sorcerer's Stone","J. K. Rowling",20.21,"Science"),
23 ("The Psychology of Money","Morgan Housel",17.43,"History");
24
```

	book_id	title	author	price	gener
▶	1	The Alchemist	Paulo Coelho	12	Science
	2	Atomic Habits	James Clear	18.99	History
	3	1984	George Orwell	10	Biography
	4	To Kill a Mockingbird	Harper Lee	14.45	History
	5	The Hobbit	J. R. R. Tolkien	15	History
	6	Rich Dad Poor Dad	Rich Dad Poor Dad	11.2	Science
	7	The Great Gatsby	F. Scott Fitzgerald	16.44	Biography
	8	Think and Grow Rich	Napoleon Hill	12.99	Biography
	9	Harry Potter and the Sorcerer's Stone	J. K. Rowling	20.21	Science
	10	The Psychology of Money	Morgan Housel	17.43	History
●	NULL	NULL	NULL	NULL	NULL

3. Retrieve all books from the library table.

	book_id	title	author	price	gener
▶	1	The Alchemist	Paulo Coelho	12	Science
	2	Atomic Habits	James Clear	18.99	History
	3	1984	George Orwell	10	Biography
	4	To Kill a Mockingbird	Harper Lee	14.45	History
	5	The Hobbit	J. R. R. Tolkien	15	History
	6	Rich Dad Poor Dad	Rich Dad Poor Dad	11.2	Science
	7	The Great Gatsby	F. Scott Fitzgerald	16.44	Biography
	8	Think and Grow Rich	Napoleon Hill	12.99	Biography
	9	Harry Potter and the Sorcerer's Stone	J. K. Rowling	20.21	Science
	10	The Psychology of Money	Morgan Housel	17.43	History
●	NULL	NULL	NULL	NULL	NULL

4. Retrieve only title and author columns

24

25 • `select title,author from library ;`

Result Grid | Filter Rows: | Export: | Wrap

	title	author
▶	The Alchemist	Paulo Coelho
	Atomic Habits	James Clear
	1984	George Orwell
	To Kill a Mockingbird	Harper Lee
	The Hobbit	J. R. R. Tolkien
	Rich Dad Poor Dad	Rich Dad Poor Dad
	The Great Gatsby	F. Scott Fitzgerald
	Think and Grow Rich	Napoleon Hill
	Harry Potter and the Sorcerer's Stone	J. K. Rowling
	The Psychology of Money	Morgan Housel

5. Retrieve books where genre is 'Science'

25 • `select * from library where gener="Science";`

Result Grid | Filter Rows: | Edit: | Export/Import:

	book_id	title	author	price	gener
▶	1	The Alchemist	Paulo Coelho	12	Science
	6	Rich Dad Poor Dad	Rich Dad Poor Dad	11.2	Science
	9	Harry Potter and the Sorcerer's Stone	J. K. Rowling	20.21	Science
*	NULL	NULL	NULL	NULL	NULL

6. Update the price of 'Harry Potter' to 699.99.

```
28 • update library set price=699.99 where title="Harry Potter";
29 • select * from library ;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap C

	book_id	title	author	price	gener
▶	1	The Alchemist	Paulo Coelho	12	Science
	2	Harry Potter	James Clear	699.99	History
	3	1984	George Orwell	10	Biography
	4	To Kill a Mockingbird	Harper Lee	14.45	History
	5	The Hobbit	J. R. R. Tolkien	15	History
	6	Rich Dad Poor Dad	Rich Dad Poor Dad	11.2	Science
	7	The Great Gatsby	F. Scott Fitzgerald	16.44	Biography
	8	Think and Grow Rich	Napoleon Hill	12.99	Biography
	9	Harry Potter and the Sorcerer's Stone	J. K. Rowling	20.21	Science
	10	The Psychology of Money	Morgan Housel	17.43	History
*	NULL	NULL	NULL	NULL	NULL

7. Update the author of book_id 3 to 'Ram kc'.

```
28 • update library set author="Ram kc" where book_id=3;
29 • select * from library ;
```

Result Grid | Filter Rows: | Edit: | Export/Import: |

	book_id	title	author	price	gener
▶	1	The Alchemist	Paulo Coelho	12	Science
	2	Harry Potter	James Clear	699.99	History
	3	1984	Ram kc	10	Biography
	4	To Kill a Mockingbird	Harper Lee	14.45	History
	5	The Hobbit	J. R. R. Tolkien	15	History
	6	Rich Dad Poor Dad	Rich Dad Poor Dad	11.2	Science
	7	The Great Gatsby	F. Scott Fitzgerald	16.44	Biography
	8	Think and Grow Rich	Napoleon Hill	12.99	Biography
	9	Harry Potter and the Sorcerer's Stone	J. K. Rowling	20.21	Science
	10	The Psychology of Money	Morgan Housel	17.43	History
*	NULL	NULL	NULL	NULL	NULL

8. Delete the book where book id = 4.

```
29 • select * from library;
30 • delete from library where book_id=4;
```

Result Grid | Filter Rows: | Edit: | Export/Import:

	book_id	title	author	price	gener
▶	1	The Alchemist	Paulo Coelho	12	Science
	2	Harry Potter	James Clear	699.99	History
	3	1984	Ram kc	10	Biography
	5	The Hobbit	J. R. R. Tolkien	15	History
	6	Rich Dad Poor Dad	Rich Dad Poor Dad	11.2	Science
	7	The Great Gatsby	F. Scott Fitzgerald	16.44	Biography
	8	Think and Grow Rich	Napoleon Hill	12.99	Biography
	9	Harry Potter and the Sorcerer's Stone	J. K. Rowling	20.21	Science
	10	The Psychology of Money	Morgan Housel	17.43	History
*	NULL	NULL	NULL	NULL	NULL

9. Delete all books where price < 600.

```
30 • delete from library where price<600;
31 • select * from library;
```

Result Grid | Filter Rows: | Edit: | Export/Import:

	book_id	title	author	price	gener
▶	2	Harry Potter	James Clear	699.99	History
*	NULL	NULL	NULL	NULL	NULL

10. Create a table called employees with:

emp_id, name not null, email unique, phone unique, department default value 'general'.

```
33 • create table employee(  
34     emp_id int primary key auto_increment,  
35     name varchar(30) NOT NULL,  
36     email varchar(20) unique,  
37     phone int unique,  
38     department varchar(30) default "general"  
39 );  
40
```

Result Grid					
Filter Rows:					
	emp_id	name	email	phone	department
*	NULL	NULL	NULL	NULL	NULL

11. Create a table called products with:

product_id, product_name, stock default value must be 0, price, must be greater than 0, Status available, out of stock and default value must be available.

```
41 • CREATE TABLE products(  
42     product_id INT PRIMARY KEY auto_increment,  
43     product_name VARCHAR(30),  
44     stock int default 0,  
45     price float ,  
46     CONSTRAINT chk_price CHECK (price > 0),  
47     status enum("Available","out of stock") default "Available"  
48 );  
49 );  
50 • select * from products;
```

Result Grid

Filter Rows:

Edit:

	product_id	product_name	stock	price	status
▶	1	Laptop	22	223	Available
	2	Mobile	40	23	Available
	3	PC	10	23	Available
⬢	NULL	NULL	NULL	NULL	NULL

12. Query to delete all records of products table but keep the table structure.

```
49  );
50  •  select * from products;
51
52  •  truncate table products;
53
```

Result Grid | Filter Rows: | Edit:

	product_id	product_name	stock	price	status
*	NULL	NULL	NULL	NULL	NULL

13. Add a new column salary (DECIMAL 10,2) to the table.

```
40  •  alter table employee add salary decimal(10,2) after email;
41  •  select * from employee;
```

Result Grid | Filter Rows: | Edit: | Export/Import:

	emp_id	name	email	salary	phone	department
*	NULL	NULL	NULL	NULL	NULL	NULL

14. Rename the column phone to mobile_number

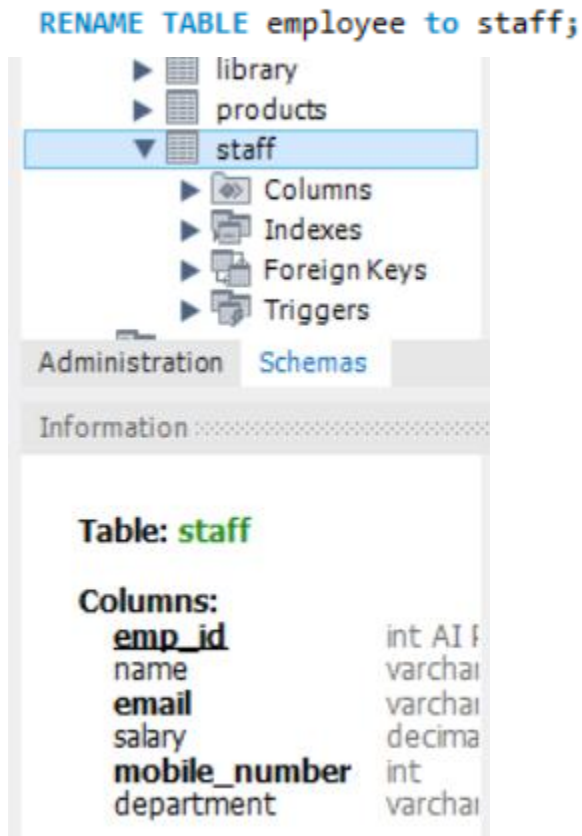
```
40  •  alter table employee add salary decimal(10,2) after email;
41  •  alter table employee rename column phone to mobile_number;
42  •  select * from employee;
```

Result Grid | Filter Rows: | Edit: | Export/Import:

	emp_id	name	email	salary	mobile_number	department
*	NULL	NULL	NULL	NULL	NULL	NULL

15. Rename the table employee to staff.

```
RENAME TABLE employee to staff;
```

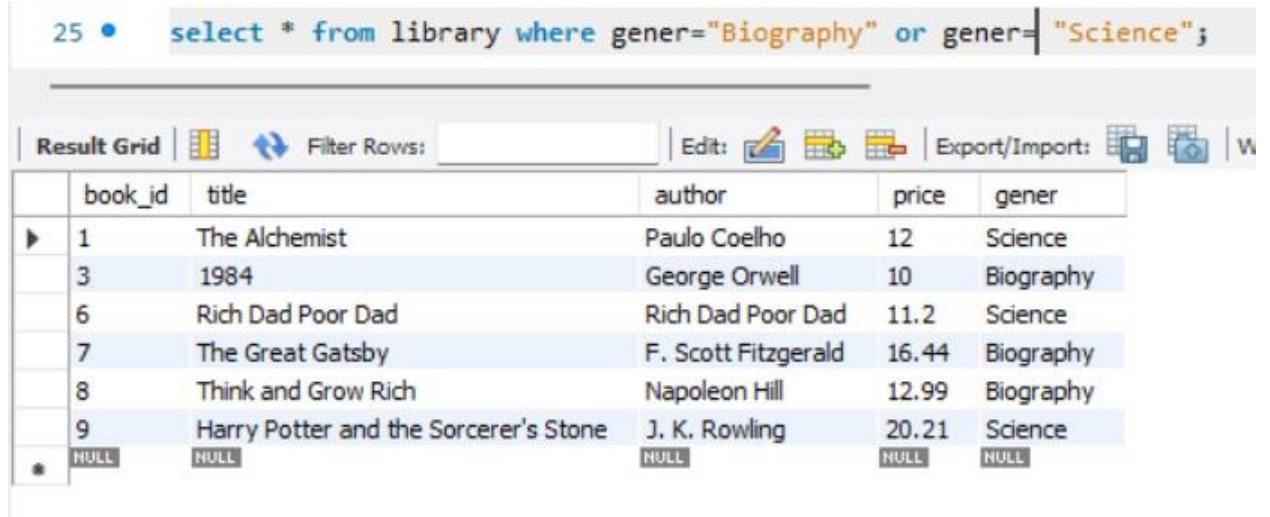


The screenshot shows a database management interface. At the top, the SQL command `RENAME TABLE employee to staff;` is entered. Below it, a tree view shows the database structure with 'library', 'products', and 'staff' tables. The 'staff' table is selected, and its properties are shown in the 'Schemas' tab. The table 'staff' has the following columns:

Column Name	Data Type
emp_id	int AI
name	varchar
email	varchar
salary	decima
mobile_number	int
department	varchar

16. Retrieve all books where genre is either 'Biography' or 'Science'.

```
25 • select * from library where gener="Biography" or gener="Science";
```



The screenshot shows the same database management interface. The SQL command `select * from library where gener="Biography" or gener="Science";` is entered. Below it, the 'Result Grid' tab is selected, showing the results of the query. The result grid has the following columns: book_id, title, author, price, and gener. The results are as follows:

book_id	title	author	price	gener
1	The Alchemist	Paulo Coelho	12	Science
3	1984	George Orwell	10	Biography
6	Rich Dad Poor Dad	Rich Dad Poor Dad	11.2	Science
7	The Great Gatsby	F. Scott Fitzgerald	16.44	Biography
8	Think and Grow Rich	Napoleon Hill	12.99	Biography
9	Harry Potter and the Sorcerer's Stone	J. K. Rowling	20.21	Science
*	NULL	NULL	NULL	NULL

17. Retrieve all books where book_id is IN (1, 3, 5)

26 • `select * from library where book_id IN (1,3,5);`

Result Grid

Filter Rows:

Edit:   

Export

	book_id	title	author	price	gener
▶	1	The Alchemist	Paulo Coelho	12	Science
	3	1984	George Orwell	10	Biography
	5	The Hobbit	J. R. R. Tolkien	15	History
✱	NULL	NULL	NULL	NULL	NULL

18. Retrieve all books where price is between 500 and 800

```
27 • select * from library where price BETWEEN 500 AND 800;
```

Result Grid		 Filter Rows:	Edit: 			Export/Import: 	
	book_id	title	author	price	gener		
▶	2	Atomic Habits	James Clear	501.99	History		
	3	1984	George Orwell	600	Biography		
	4	To Kill a Mockingbird	Harper Lee	550	History		
	5	The Hobbit	J. R. R. Tolkien	599.99	History		
	6	Rich Dad Poor Dad	Rich Dad Poor Dad	777.2	Science		
	7	The Great Gatsby	F. Scott Fitzgerald	799.44	Biography		
	8	Think and Grow Rich	Napoleon Hill	699.99	Biography		
	9	Harry Potter and the Sorcerer's Stone	J. K. Rowling	666.66	Science		
•	NULL	NULL	NULL	NULL	NULL		

19. Retrieve all orders placed between '2024-01-01' and '2024-12-31'

79

80 • `select * from orders where order_date between '2024-01-01' and '2024-12-31';`

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Conte

	product_id	name	price	qty	order_date
▶	1	Laptop	55000.00	2	2024-01-15
	2	Mouse	1500.00	5	2024-03-22
	3	Keyboard	2500.00	3	2024-05-10
	4	Monitor	18000.00	1	2024-07-04
	5	Headphones	3500.00	4	2024-09-18
	6	Webcam	2200.00	2	2024-11-30
*	NULL	NULL	NULL	NULL	NULL

20. Write a query using CASE to:

If department is 'IT' increase salary by 20%

If department is 'HR' increase salary by 15%

If department is 'Finance' increase salary by 10%

All others increase salary by 5%

```
SELECT * FROM employee;  
  
INSERT INTO employee VALUES  
(1, 'Nischal', 'IT', 50000.00),  
(2, 'Aarav', 'HR', 40000.00),  
(3, 'Sneha', 'Finance', 45000.00),  
(4, 'Rohan', 'Sales', 35000.00),  
(5, 'Priya', 'IT', 60000.00);
```

```
98 • UPDATE employee  
99 SET  
100 salary = CASE  
101 WHEN department = 'IT' THEN salary*1.2  
102 WHEN department='HR' THEN salary*1.15  
103 WHEN department='Finance' THEN salary*1.1  
104 ELSE salary*1.05  
105 END;
```

Result Grid				
Filter Rows:				
	emp_id	emp_name	department	salary
▶	1	Nischal	IT	60000.00
	2	Aarav	HR	46000.00
	3	Sneha	Finance	49500.00
	4	Rohan	Sales	36750.00
	5	Priya	IT	72000.00
*	NULL	NULL	NULL	NULL